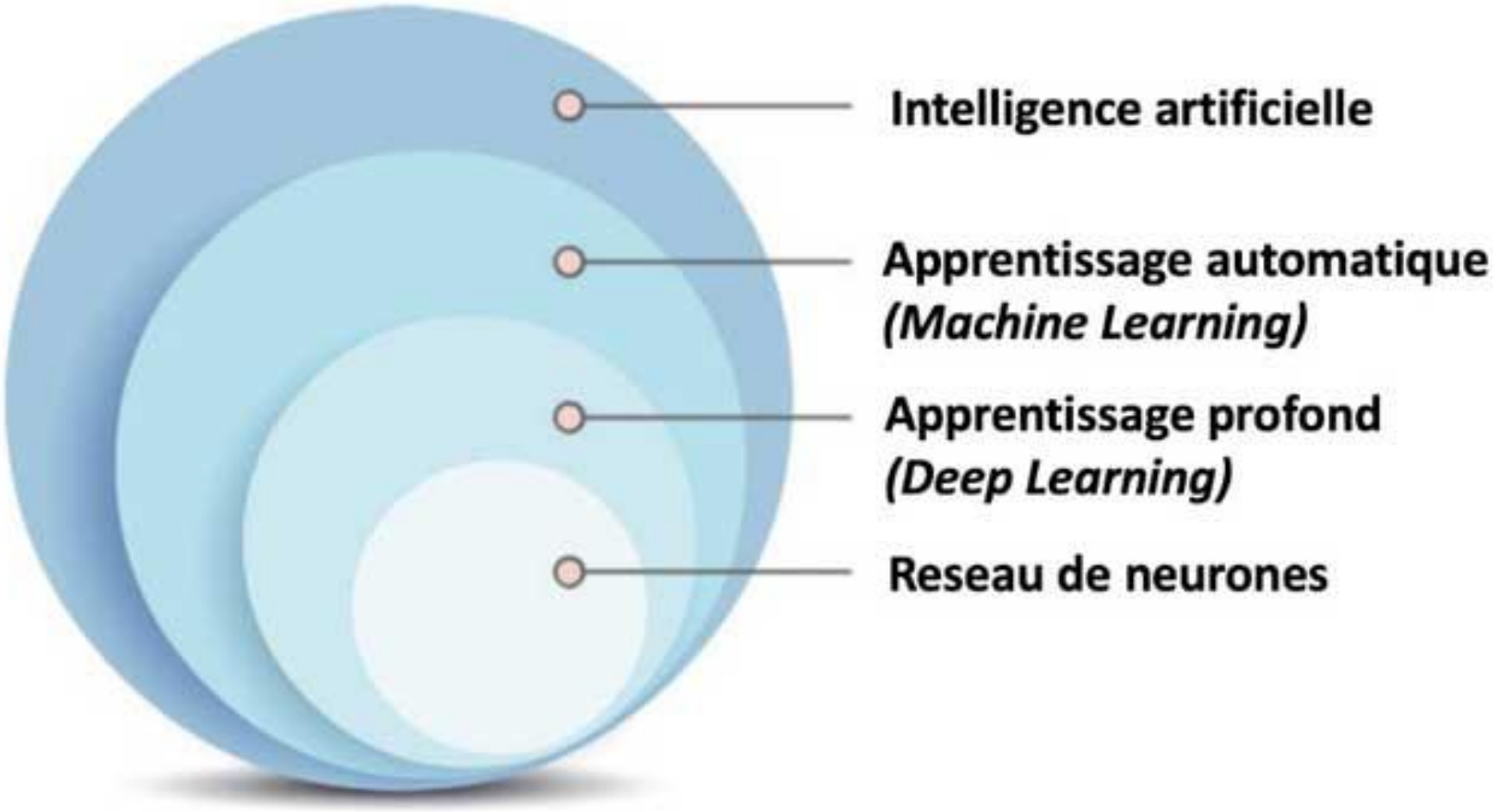
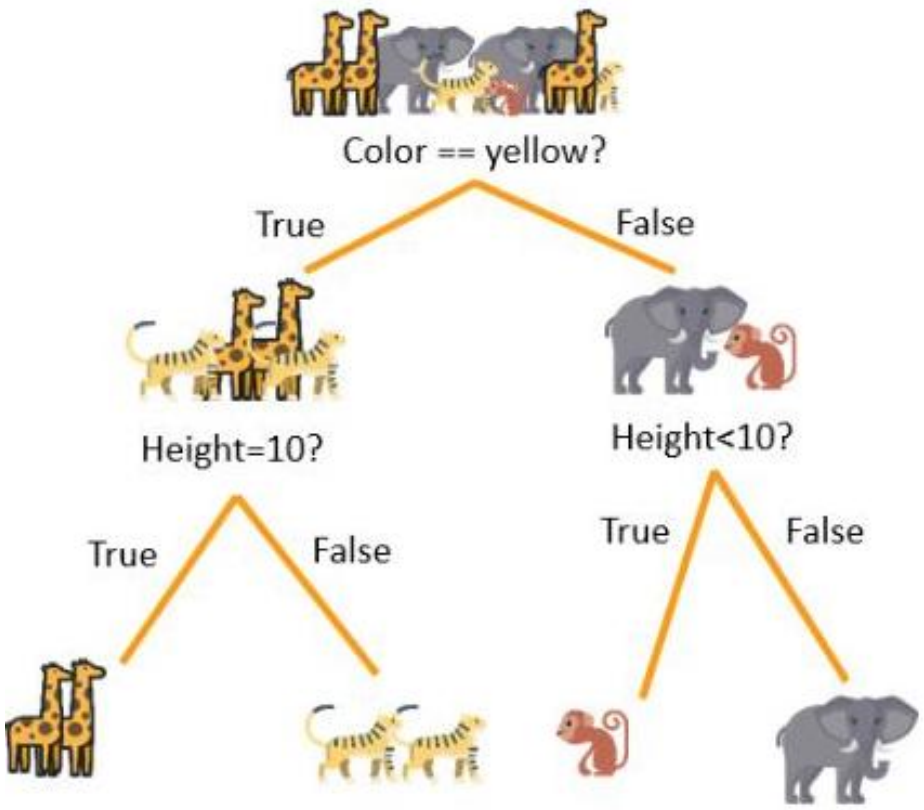
Formation 8 :

Introduction au Deep Learning

KRYPTOSPHERE®


Intelligence artificielle
Apprentissage automatique
(Machine Learning)
Apprentissage profond
(Deep Learning)
Reseau de neurones

Animal	Height (cm)	Weight (kg)	Color	Lifespan (years)	Diet	Habitat	Predators	Average Speed (km/h)	Countries Found	Conservation Status	Family	Gestation Period (days)	Top Speed (km/h)
Aardvark	105-130	40-65	Grey	20-30	Insectivore	Savannas, Grasslands	Lions, Hyenas	40	Africa	Least Concern	Orycteropodidae	210-240	40
Aardwolf	40-50	8-14	Yellow-brown	10-12	Insectivore	Grasslands, Savannas	Lions, Leopards	24-30	Eastern and Southern Africa	Least Concern	Hyaenidae	90	40
African Elephant	270-310	2700-6000	Grey	60-70	Herbivore	Savannah, Forest	Lions, Hyenas	25	Africa	Vulnerable	Elephantidae	640-660	40
African Lion	80-110	120-250	Tan	10-14	Carnivore	Grasslands, Savannas	Hyenas, Crocodiles	58	Africa	Vulnerable	Felidae	98-105	80
African Wild Dog	75-80	18-36	Multicolored	10-12	Carnivore	Savannahs	Lions, Hyenas	56	Sub-Saharan Africa	Endangered	Canidae	70	56



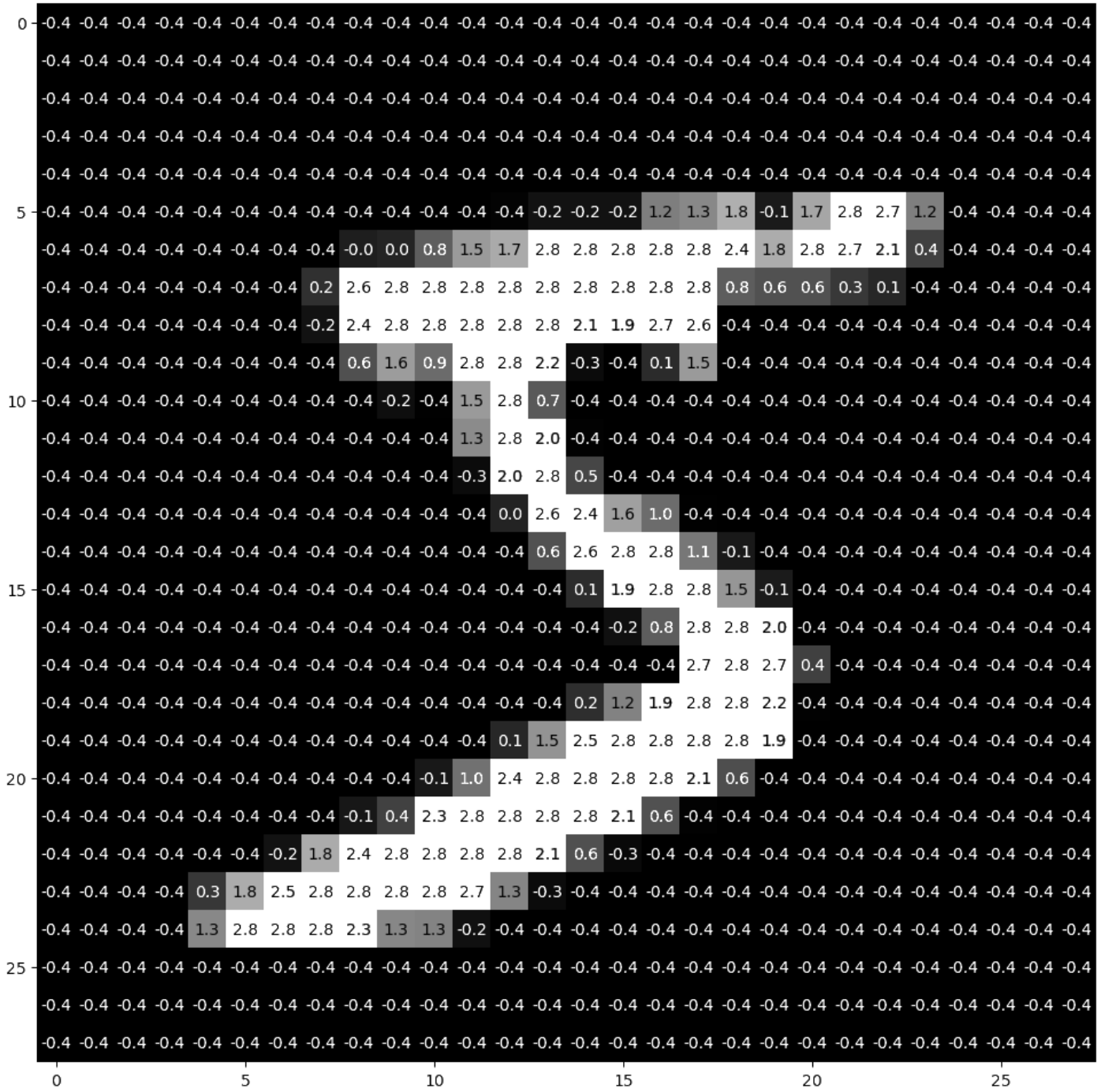
Rappels sur le ML Classique

Intérêt du Deep Learning

Cat

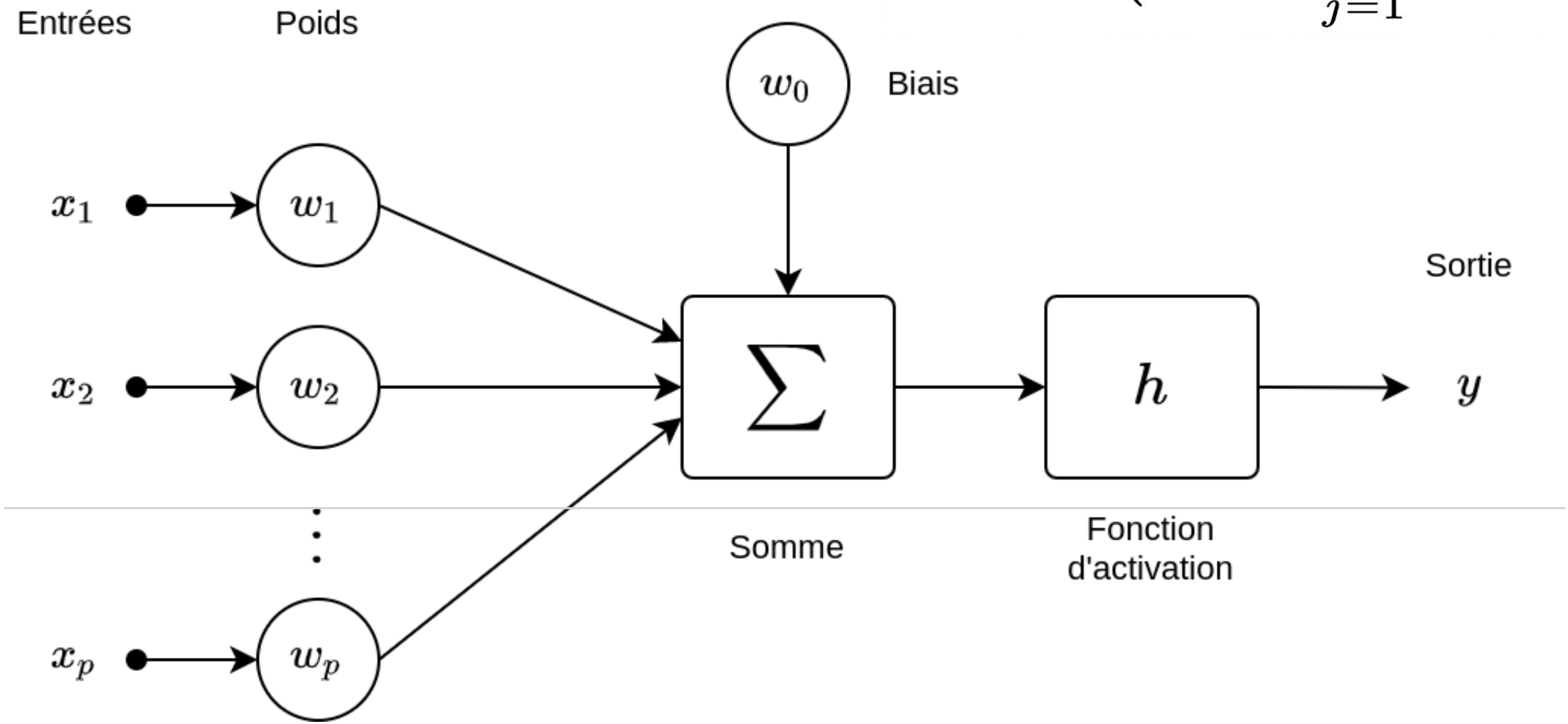
Dog

Exemple Classique : Mnist

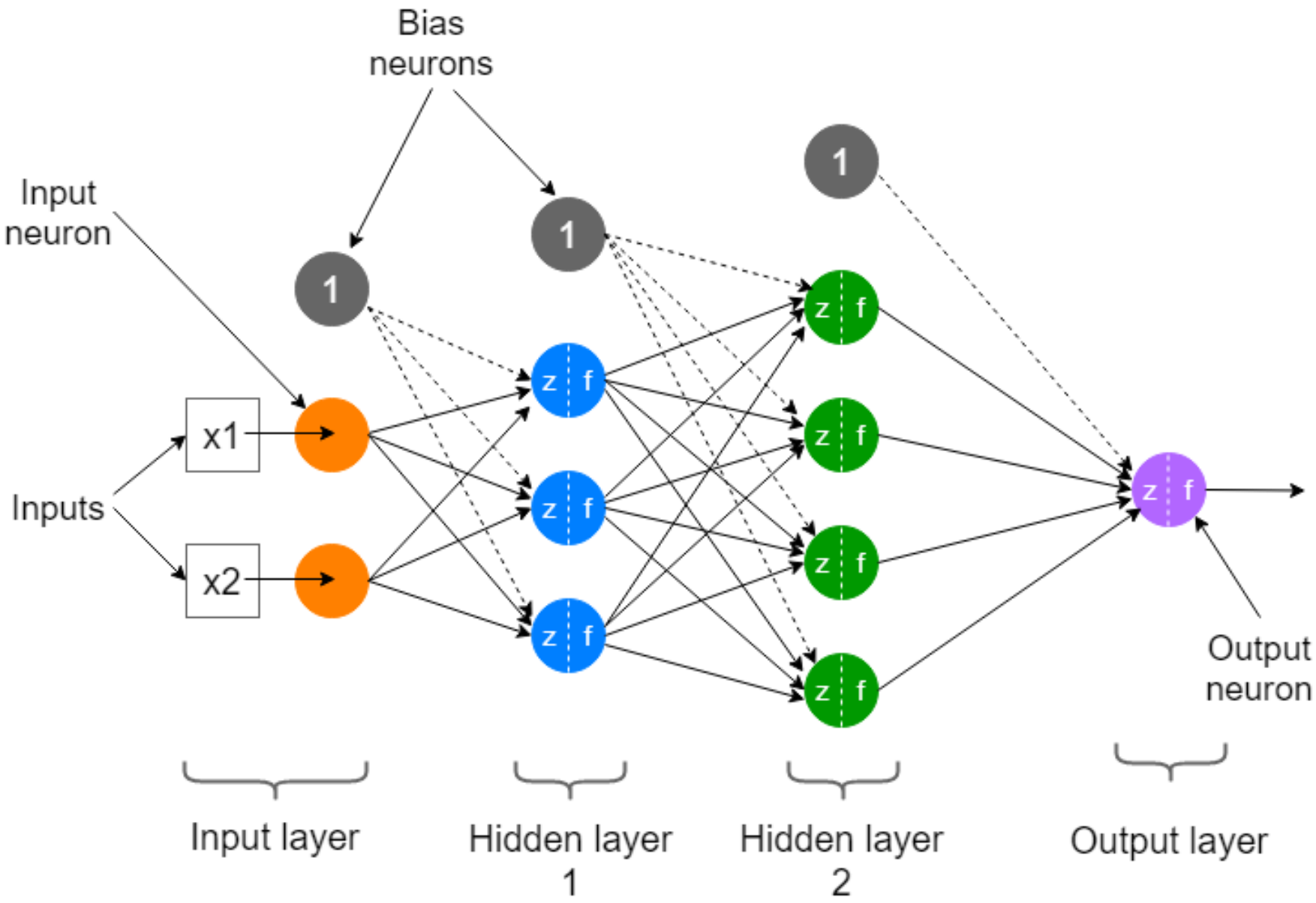


Un neurone (perceptron)

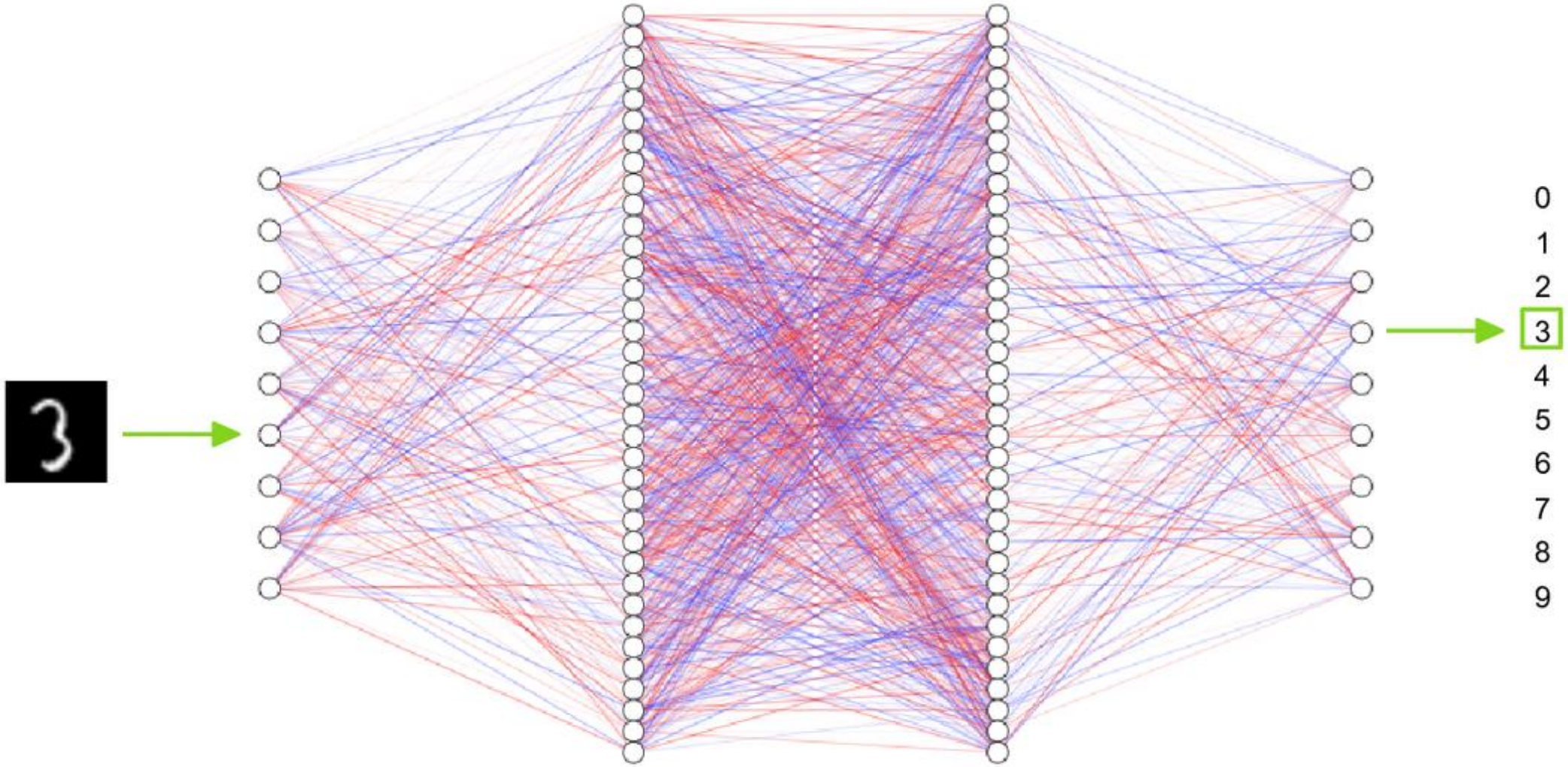
$$\mathbf{y} = h\left(\omega_0 + \sum_{j=1}^{p} \omega_j x_j\right)$$



- Réseau de Neurone

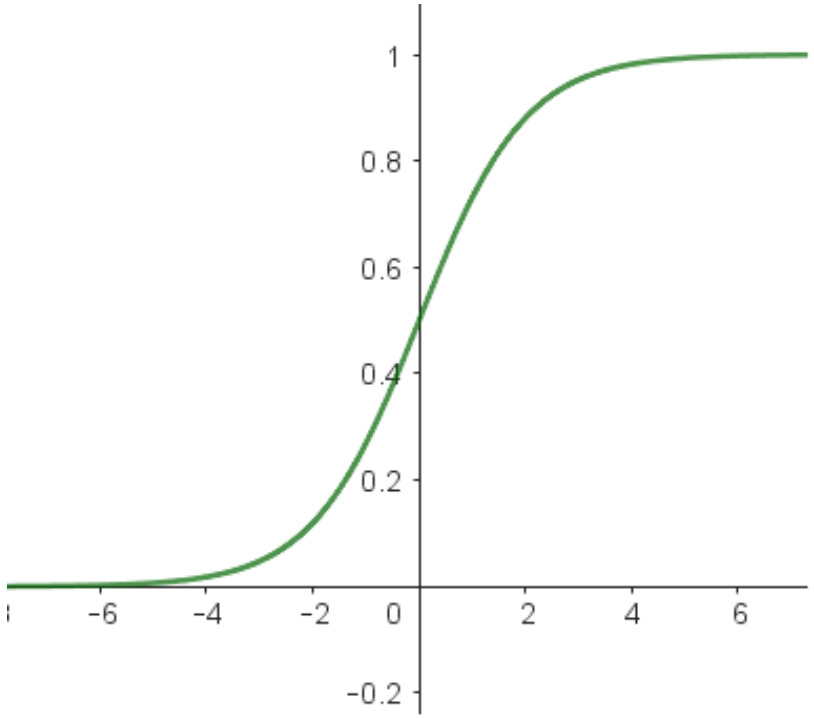


Retour sur Mnist



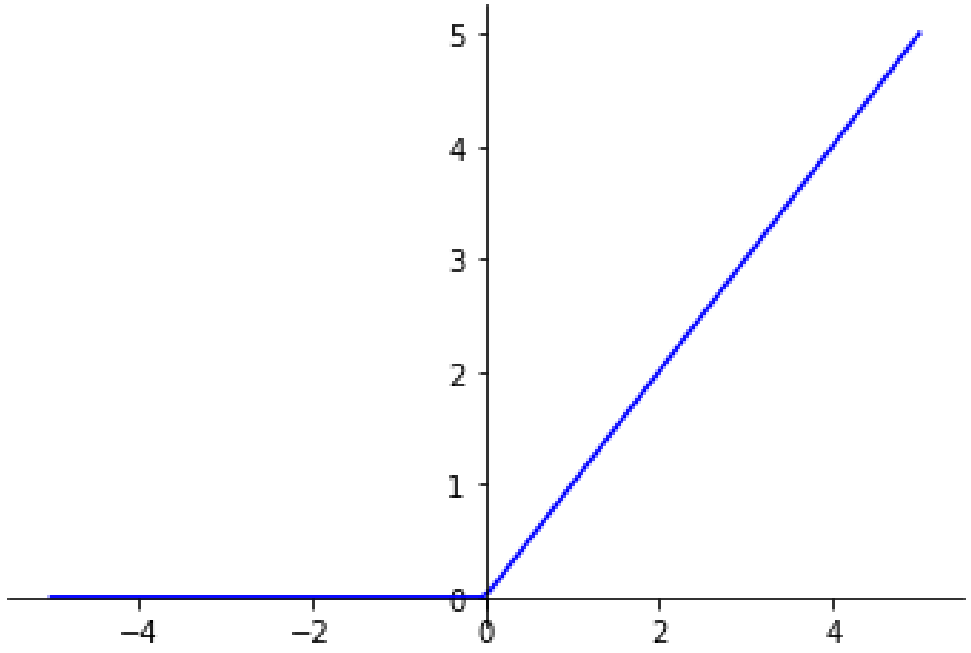
Les principales fonctions d'activation

Sigmoide



$$\phi(z) = \frac{1}{1 + e^{-z}}$$

Relu

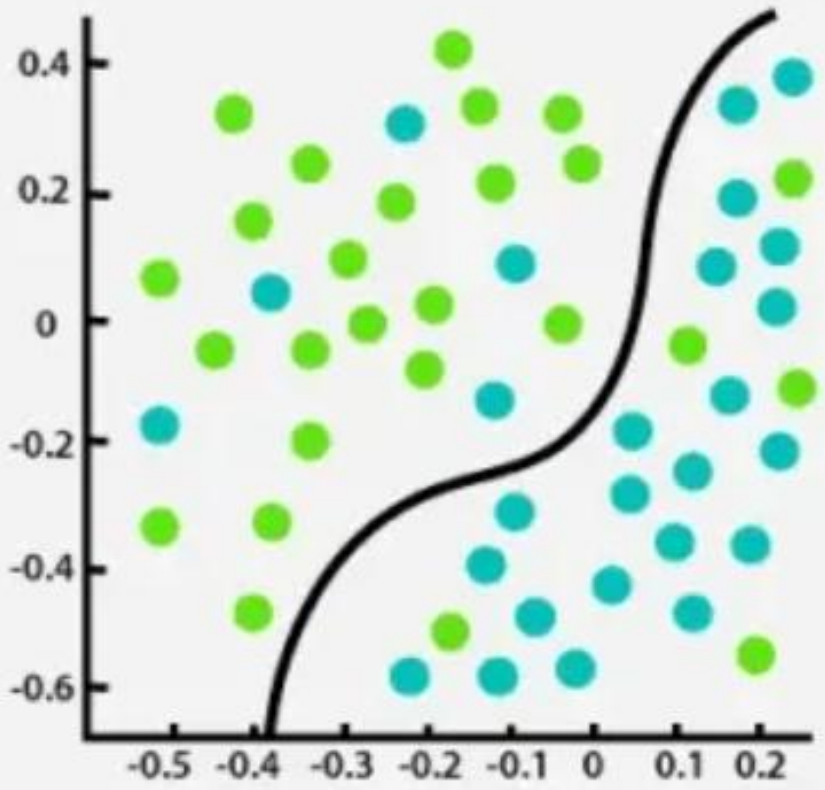


$$f(x) = \max(0,x)$$

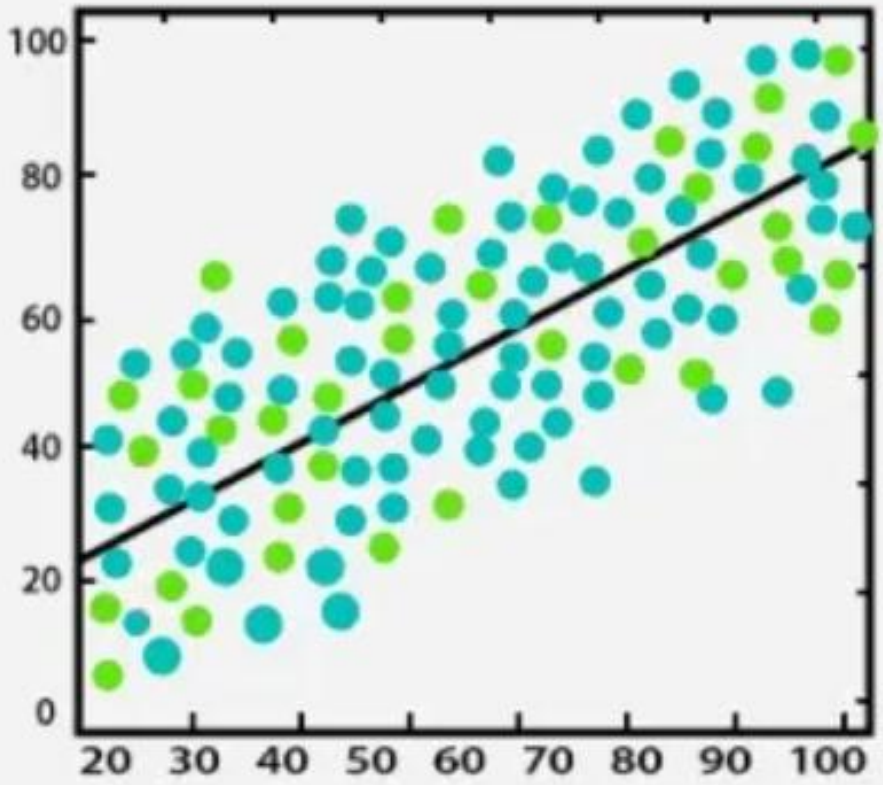
SoftMax

$$\frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$$

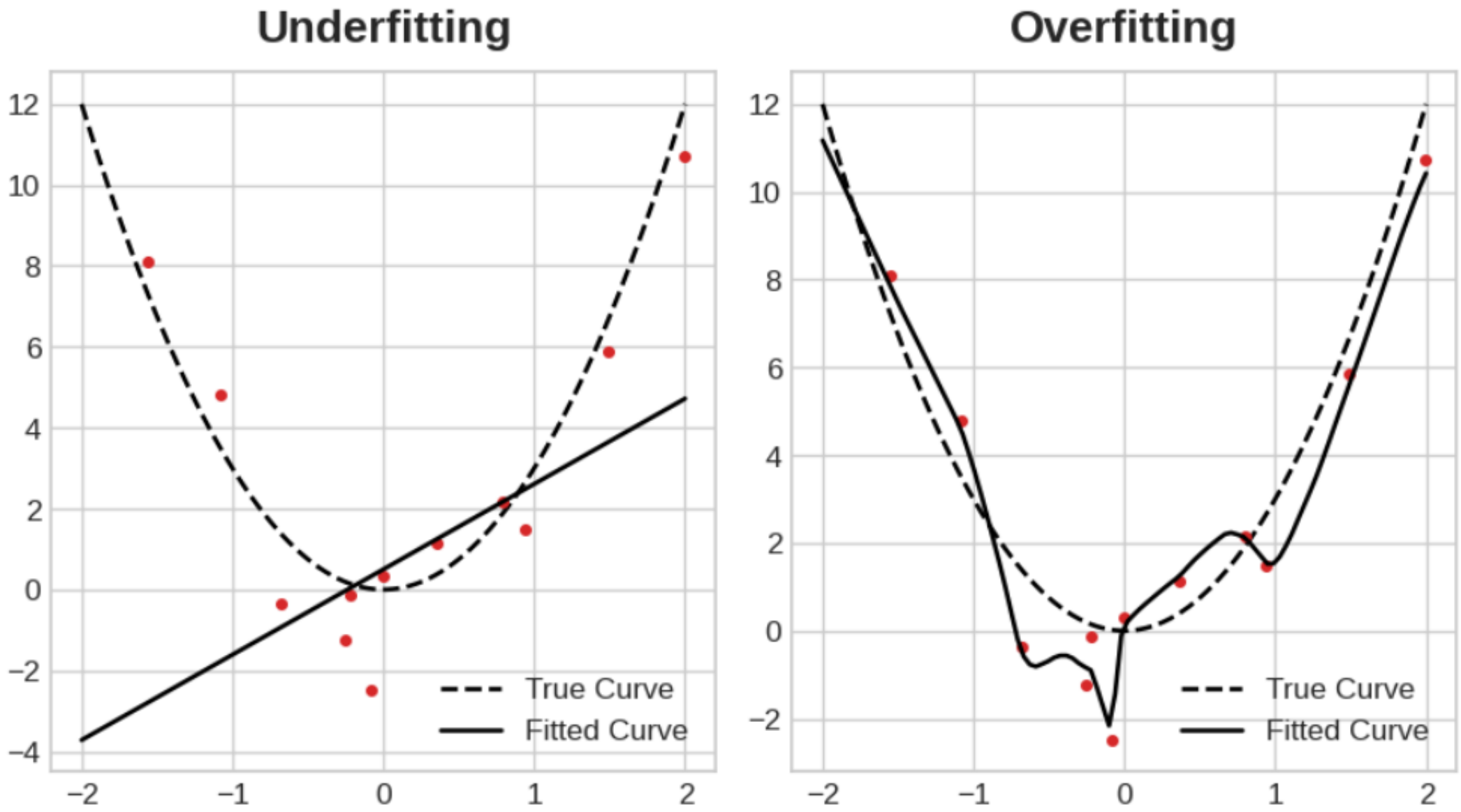
Quelques Rappels de ML



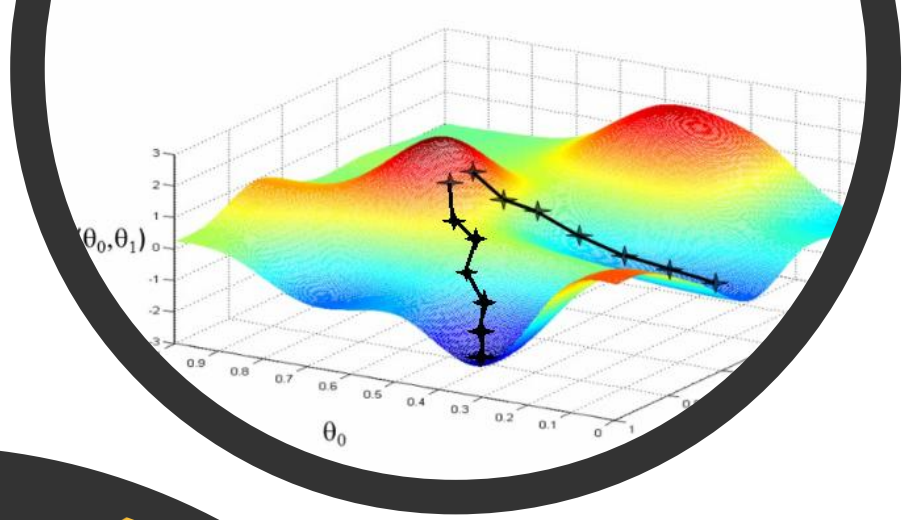
Classification

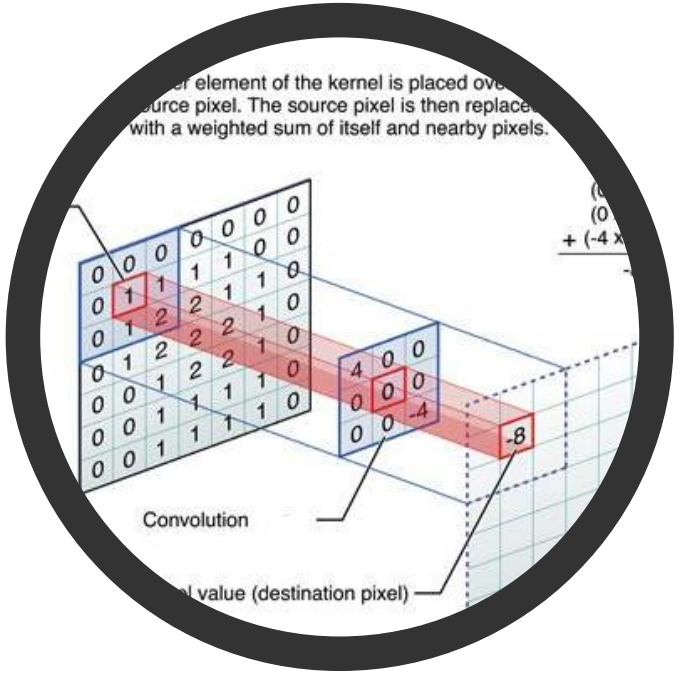
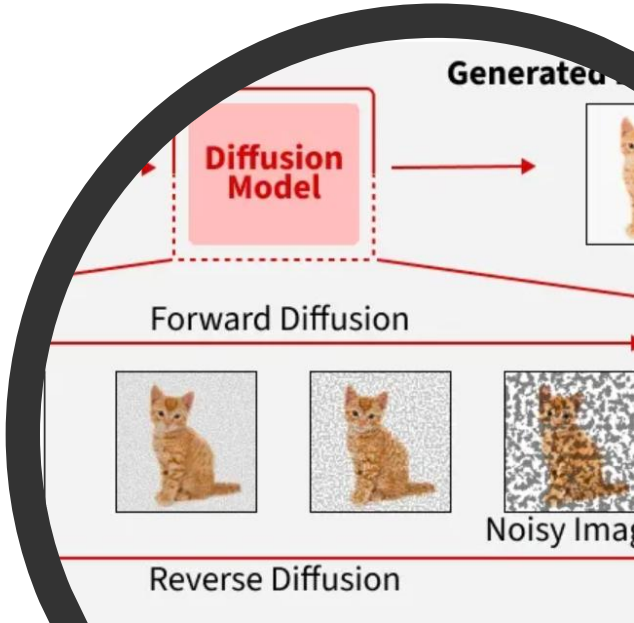


Regression



Underfitting and overfitting.





Au programme des prochaines séances :